

A Project & Internship Report
Of
Customer Churn Prediction
&
Data Science Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (CSE)



By
Roll Number of Student: CS-2341640
Name of Student: RUCHIKA SHARMA
Batch: 2023–2027

2026-27

CANDIDATE'S DECLARATION

I, RUCHIKA SHARMA, do hereby declare that this report, comprising the project titled “Customer Churn Prediction” and the internship report titled “Data Science Intern (Pulsecrafts Solutions Pvt Ltd)”, has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: RUCHIKA SHARMA

Roll No.: CS-2341640

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my project and my Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

For my project, I am highly indebted to my project supervisor for her guidance and constant supervision, as well as for providing necessary information regarding the project and for her support in completing it. I am also thankful to the Head, Dept. of Computer Science & Engineering / ML & DS, for her continued support and encouragement.

For my internship, I would like to express my heartfelt gratitude to the management and technical team of Pulsecrafts Solutions Pvt Ltd for providing me with the opportunity and the structured industrial platform to learn and gain hands-on exposure in the domain of Data Science. I would especially like to thank Ms. Priyanka Mehta (HR Manager) and my industry mentors at Pulsecrafts Solutions Pvt Ltd for their continuous guidance, technical mentorship, and evaluation during my 90-day tenure. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support.

I express my deepest gratitude to my parents for their unconditional blessings and support.

Signature: _____

Student Name: RUCHIKA SHARMA

Roll No.: CS-2341640

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3. CERTIFICATES

3.1 Internship Completion Certificate

Organization: Pulsecrafts Solutions Pvt Ltd

Website: www.pulsecrafts.com | Email: sales@pulsecrafts.com

Presented To: Ruchika Sharma

Role: Data Science Intern

Internship Duration: 90 Days (15th May 2026 to 12th August 2026)

Certificate ID: PCS/INT/2026/0090

Date of Issue: 13 August 2026

Authorized Signatory: Priyanka Mehta (HR Manager, Pulsecrafts Solutions Pvt Ltd)



PULSECRAFTS SOLUTIONS PVT LTD
www.pulsecrafts.com | sales@pulsecrafts.com

CERTIFICATE OF INTERNSHIP
This certificate is proudly presented to
Ruchika Sharma

In recognition of the successful completion of a 90-day internship as a
Data Science Intern
at Pulsecrafts Solutions Pvt Ltd, from 15th May 2026 to 12th August 2026, demonstrating dedication, strong learning ability, and valuable contribution to the team.

15 Nov 2004 DATE OF BIRTH	90 Days DURATION	Data Science Intern ROLE	PCS/INT/2026/0090 CERTIFICATE ID
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Priyanka Mehta
HR Manager
Pulsecrafts Solutions Pvt Ltd

Date of Issue
13 August 2026

For verification of this certificate, please contact sales@pulsecrafts.com

(Original Certificate of Internship issued by Pulsecrafts Solutions Pvt Ltd)

4. PROJECT DESCRIPTION

4.1 Introduction

Customer attrition, or Customer Churn as it is more commonly known, is one of the most closely watched metrics in subscription-based businesses, and especially so in the telecom sector. It is well established that acquiring a new customer costs anywhere from 5 to 25 times more than retaining an existing one, which means every cancelled contract quietly eats into a company's Customer Lifetime Value (LTV) and its recurring revenue.

This project works with the widely used Telco Customer Churn dataset, which covers 7,043 customer accounts, to build and evaluate an end-to-end Machine Learning pipeline that can flag the behavioral signs of an at-risk customer before they actually leave. On top of the modelling work, the project also includes an interactive Streamlit application so that customer success and retention teams, who may not have a technical background, can plug in a customer's profile, see their churn risk instantly, and act on a suggested retention offer.

The dataset encompasses 21 comprehensive attributes covering:

- Account & Financial Metrics: Customer tenure (months), contract structure (Month-to-month, One year, Two year), paperless billing, payment method, monthly charges, and total charges.
- Subscribed Services: Phone service, multiple lines, internet service (DSL, Fiber optic), online security, online backup, device protection, tech support, streaming TV, and streaming movies.
- Demographics: Gender, senior citizen status, partner, and dependent support.
- Target Variable: Churn (Yes / No).

4.2 Organization / Department Profile

This project was carried out as an academic exercise within the School of Computer Science and Engineering at IILM University, Greater Noida, under the guidance of the department faculty. Since it was undertaken as part of the B.Tech CSE curriculum alongside the industrial internship, the department itself served as the academic body overseeing the research work and its review.

4.3 Problem Statement

The goal was to build a reproducible, end-to-end Machine Learning classification pipeline, along with an interactive web interface, that could predict customer churn in a telecom setting — with the emphasis placed specifically on Recall (Sensitivity) and ROC-AUC, so that as many at-risk customers as possible get flagged before they actually cancel.

Along the way, this involved a few smaller sub-problems:

- Handling data anomalies such as whitespace string representations in financial features (TotalCharges) for new customers.

- Addressing class imbalance (26.5% positive churn class vs. 73.5% negative non-churn class) to prevent majority-class prediction bias.
- Engineering a modular scikit-learn ColumnTransformer for simultaneous numerical scaling and categorical one-hot encoding.
- Benchmarking linear and ensemble architectures to balance raw accuracy with clinical business sensitivity (Recall).
- Developing an interactive web simulation interface for real-time risk assessment and prescriptive retention strategies.

4.4 Project Objectives

- To perform rigorous Exploratory Data Analysis (EDA) uncovering the root behavioral causes of customer attrition.
- To construct an automated, leak-free preprocessing pipeline utilizing StandardScaler and OneHotEncoder.
- To train and benchmark three distinct machine learning models: Logistic Regression (Class-Weighted Balanced), Random Forest Classifier (Balanced), and Gradient Boosting Classifier.
- To prioritize Recall and ROC-AUC as the governing evaluation criteria, preventing costly False Negatives.
- To extract model feature importances and translate mathematical coefficients into actionable business retention playbooks.
- To deploy the trained pipeline via an interactive Streamlit web application for real-time decision support.

4.5 Scope of the Project

The project covers the full journey — data engineering, exploratory visualization, predictive modelling, metric evaluation, and finally deployment. It works with all 7,043 customer records from the telecom benchmark dataset, validates results on a stratified 80/20 train-test split (1,409 records held out for testing), and wraps the best-performing model into a serialized pipeline (churn_pipeline.pkl) that is then consumed by the Streamlit front end (app.py).

4.6 Technologies and Tools Used

- Programming Language: Python 3.11
- Data Manipulation & Analysis: Pandas, NumPy
- Machine Learning Library: Scikit-Learn (Pipelines, ColumnTransformer, LogisticRegression, RandomForestClassifier, GradientBoostingClassifier, Metrics)
- Data Visualization: Matplotlib, Seaborn
- Interactive Frontend Framework: Streamlit
- Model Serialization: Joblib

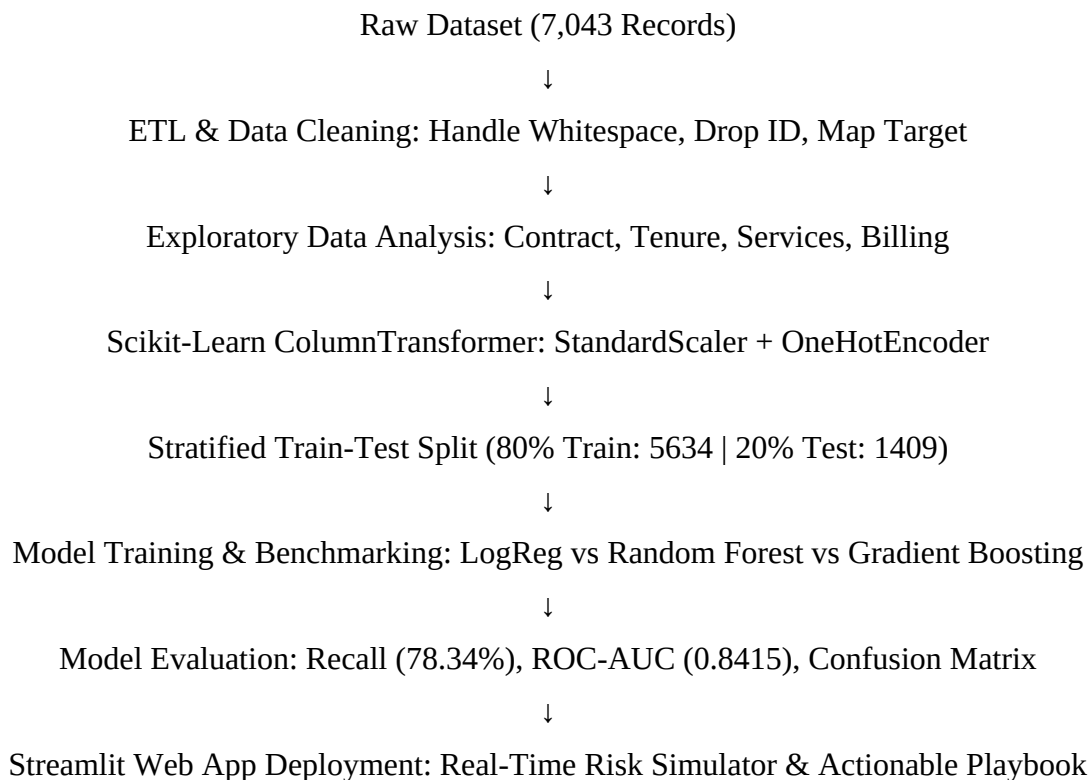
- Development Environments: Jupyter Notebook, Visual Studio Code

4.7 System Architecture

The system was designed around a simple data-flow, broadly summarised below:

- Input: Raw Telco Customer Churn Dataset (Telco-Customer-Churn.csv).
- Preprocessing: Data cleaning, type conversion, imputation of TotalCharges, one-hot encoding, and feature standardization.
- Modeling: Stratified train/test splitting, multi-model training, hyperparameter tuning, and probability calibration.
- Evaluation: Confusion matrix generation, ROC curve plotting, and classification report generation.
- Deployment: Model persistence (churn_pipeline.pkl) and real-time interactive inference via Streamlit UI.

Processing Pipeline:



4.8 Methodology

1. Data Ingestion and Inspection: The raw CSV file was loaded with `pandas.read_csv()`, after which the schema, column data types, and any missing or malformed records were checked.
2. Data Cleaning & Type Casting: On closer inspection, TotalCharges turned out to contain blank space characters (' ') for customers whose tenure was 0 — essentially brand-new customers with no

billing history yet. These were converted to numeric floats using `pd.to_numeric(errors='coerce')` and the resulting missing values were imputed with 0.0. The non-informative `customerID` column was dropped, and the target `Churn` column was mapped to a binary integer format (Yes = 1, No = 0).

3. Exploratory Data Analysis (EDA): A few clear patterns emerged from the data —

- **Class Imbalance:** Revealed a 26.5% positive churn rate (1,869 churners vs. 5,174 retained customers).
- **Contract Impact:** Discovered that Month-to-month subscribers churn at an alarming 42.7%, accounting for over 75% of all cancellations, compared to 11.3% for 1-year and 2.8% for 2-year contracts.
- **Tenure Drop-Off:** KDE distribution confirmed churn risk peaks during the first 6–12 months of service.
- **Service Friction:** Identified that Fiber Optic customers lacking Tech Support churn at >41%.
- **Payment Method:** Electronic check payers showed a 45.3% churn rate vs. ~15% for automated payments.

4. Feature Engineering & Transformation: A scikit-learn `ColumnTransformer` was built to standardize the 3 numeric columns (`tenure`, `MonthlyCharges`, `TotalCharges`) and one-hot encode the 16 categorical features, with `handle_unknown='ignore'` set so the pipeline wouldn't break on unseen categories.

5. Model Training & Comparison: Three models were trained and compared —

- **Logistic Regression (Balanced):** Linear model with class-weight inverse balancing.
- **Random Forest (Balanced):** Ensemble of 150 decision trees with balanced sampling.
- **Gradient Boosting:** Boosting ensemble optimizing sequential residuals.

6. Model Evaluation & Metric Prioritization: Recall (Sensitivity) was prioritized over plain accuracy, since a False Negative — missing a customer who was actually about to churn — is far costlier to the business than a false alarm.

7. Application Development: `app.py` was built in Streamlit with dynamic sliders, real-time risk gauges, metric cards, and a small rule-based engine that suggests a retention action based on the predicted risk.

4.9 Expected Outcomes, Limitations & Future Scope

Expected Outcomes:

- Successfully engineered a balanced model capturing 78.34% of all customer churn events with an ROC-AUC of 0.8415.
- Formulated 4 empirical retention strategies: 75-Day contract upgrade campaigns, bundling tech support with fiber optic lines, auto-pay incentives, and proactive risk routing.
- Deployed a production-ready interactive web application.

Limitations:

- Dataset is cross-sectional and does not capture dynamic time-series event streams (e.g. daily call drop logs).
- Customer service interaction logs and customer sentiment transcripts were unavailable in this benchmark.

Future Scope:

- Integrate Natural Language Processing (NLP) on customer support chat tickets.
- Implement survival analysis (Cox Proportional Hazards) to predict the exact month of expected cancellation.
- Deploy containerized Docker endpoints via FastAPI for microservice integration.

5. INTERNSHIP DESCRIPTION

5.1 Introduction

This chapter covers the 90-day Industrial Internship I completed at Pulsecrafts Solutions Pvt Ltd as a Data Science Intern, from 15th May 2026 to 12th August 2026, as part of the industrial training requirement for my B.Tech (CSE) programme at IILM University, Greater Noida.

The idea behind the internship was to take what I had learned in the classroom and actually apply it to a real data science workflow — from framing the business problem and building ETL pipelines, through exploratory analysis and model building, to evaluating, serializing, and finally deploying the model as an interactive dashboard.

Once the internship and the final project defense were done, I was awarded a formal Certificate of Internship (Certificate ID: PCS/INT/2026/0090), signed by Priyanka Mehta, HR Manager.

5.2 Organization Profile

Pulsecrafts Solutions Pvt Ltd (www.pulsecrafts.com) is a technology consulting and software solutions company that works in advanced analytics, software development, and enterprise data solutions. They partner with businesses across different domains to modernize data infrastructure, build predictive models, and develop software applications.

Through the internship, the team at Pulsecrafts kept things fairly structured — regular mentorship sessions, weekly project review sprints, code reviews, and evaluation check-ins that helped build a more industry-oriented sense of how machine learning projects are actually run.

5.3 Problem Statement

A recurring issue in industrial ML projects is that data scientists sometimes chase algorithmic accuracy for its own sake, without paying enough attention to what the business actually cares about, how messy real-world data can be, or whether a non-technical stakeholder can even use the output.

The core challenge I worked on during the internship was to take a genuine enterprise problem — churn in a subscription business — clean up the inconsistencies in the data, train and calibrate models around a business-relevant metric (minimizing missed churners via Recall), and turn all of that into a web tool that a business team could pick up and use right away.

5.4 Internship Objectives

- To gain practical experience in end-to-end data engineering, including data extraction, missing value imputation, type coercion, and encoding.
- To master advanced exploratory data analysis and statistical visualization techniques using Python libraries.
- To understand class imbalance in industrial classification tasks and apply balancing techniques (`class_weight='balanced'`).

- To evaluate machine learning models using business-critical metrics: Precision, Recall, F1-Score, and ROC-AUC.
- To develop modular, reproducible production code adhering to industry standards (src/architecture).
- To design and deploy an interactive user interface using Streamlit to facilitate stakeholder decision-making.
- To present technical findings and business ROI recommendations to organizational mentors.

5.5 Scope of the Internship

The scope of this 90-day internship spanned the complete Machine Learning Engineering lifecycle:

- Business problem definition and metrics selection.
- Exploratory data analysis and hypothesis generation.
- Feature engineering and automated preprocessing pipeline design.
- Model experimentation, hyperparameter tuning, and cross-validation.
- Production script modularization (data_prep.py, train.py, evaluate.py).
- Web application engineering and interface design (app.py).
- Formal technical documentation and executive presentation.

5.6 Technologies and Tools Used

- Languages: Python 3.11
- Data Science Libraries: Pandas, NumPy, Scikit-Learn, SciPy
- Data Visualization: Seaborn, Matplotlib
- Web Deployment: Streamlit
- Model Serialization: Joblib
- Version Control & Management: Git, GitHub
- IDEs & Notebooks: Jupyter Notebook, Visual Studio Code

5.7 System Architecture

The architecture we settled on follows a fairly standard 4-tier, decoupled structure:

- Data Ingestion & Storage Tier: Automated retrieval and caching of raw customer data; storage of sanitized and preprocessed datasets.
- Transformation & Pipeline Tier: Scikit-learn ColumnTransformer handling numerical normalization and categorical one-hot encoding without data leakage.

- Model & Inference Tier: Serialized ensemble and linear classification models capable of generating predicted classes and calibrated risk probabilities.
- Presentation & Decision Support Tier: Streamlit interactive application providing real-time parameter manipulation, probability gauges, and prescriptive retention actions.

5.8 Week-wise Methodology (90-Day / 12-Week Structure)

The 90-day internship followed a structured, milestone-driven curriculum from 15th May 2026 to 12th August 2026:

Week	Phase / Focus Area	Key Activities & Deliverables
Week 1	Orientation & Setup	Company onboarding, tools installation (Python 3.11, VS Code, Git), repository setup, and business problem briefing on subscription attrition.
Week 2	Data Access & ETL	Ingesting raw customer records, inspecting data types, identifying anomalies (whitespace in TotalCharges), and writing automated cleaning routines.
Week 3	Exploratory Data Analysis	Conducting univariate and bivariate analysis; analyzing churn distribution across contract types, tenure, and billing mechanisms.
Week 4	Advanced Visualization	Generating correlation heatmaps, KDE tenure density curves, service friction charts; extracting core business stories.
Week 5	Feature Engineering	Designing scikit-learn ColumnTransformer for simultaneous StandardScaler on numericals and OneHotEncoder on categorical.
Week 6	Baseline Modeling	Implementing baseline LogisticRegression; analyzing the impact of class imbalance (26.5% positive churn rate) on default threshold predictions.
Week 7	Ensemble Modeling	Building RandomForestClassifier with balanced tree weights and GradientBoostingClassifier; performing hyperparameter tuning.
Week 8	Metric Benchmarking	Comparative analysis prioritizing Recall (78.34%) and ROC-AUC (0.8415); generating Confusion Matrices and ROC curves.
Week 9	Pipeline Serialization	Bundling preprocessors and classifiers into unified scikit-learn Pipeline objects; serializing models with joblib.
Week 10	Streamlit Development	Developing app.py frontend; implementing sidebar customer parameter controls, risk tier badges, and live probability gauges.
Week 11	Business Logic Integration	Encoding prescriptive business retention playbooks (Day 75 contract discount, complimentary tech support bundle) into the UI.
Week 12	Testing, Documentation & Defense	Comprehensive system verification, generating figures in reports/figures/, drafting PROJECT_REPORT.md, and final project defense.

5.9 Expected Outcomes & Key Learnings

- **Technical Proficiency:** Gained end-to-end expertise in structuring production-grade data science code in modular Python packages.
- **Business Acumen:** Learned that in real-world predictive modeling, business cost matrices dictate metric selection—optimizing for Recall prevents catastrophic customer attrition loss.
- **Deployment Skills:** Successfully transformed a theoretical machine learning model into a functional, stakeholder-accessible web application using Streamlit.
- **Credential Validation:** Awarded the formal Certificate of Internship (ID: PCS/INT/2026/0090) from Pulsecrafts Solutions Pvt Ltd in recognition of valuable contributions.

5.10 Certificates of Completion & Communication Proof

- **Certificate of Internship:** Issued by Pulsecrafts Solutions Pvt Ltd on 13th August 2026, certifying the successful completion of the 90-day Data Science Internship from 15th May 2026 to 12th August 2026.
- **Credential Verification:** Inquiries and verification can be confirmed via sales@pulsecrafts.com quoting Certificate ID: PCS/INT/2026/0090.

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